

PILOT CAT

GAME DEVELOPMENT DOCUMENT



Contributors: Luisa, Pascual, Roman, Vinh

Table of Contents

1. Game Concept and Story Development

1.1 Game Overview

1.2 Game Concept and Vision

1.3 Game Genre and Target Audience

1.4 Game Setting and World

1.5 Backstory and Lore

1.6 Main Story and Narrative

1.7 Player Identity and Role

1.8 Player Goals and Motivation

1.9 Intended Player Experience

2. Game Design and Gameplay Planning

2.1 Core Gameplay Loop

2.2 Gameplay Mechanics

2.3 Player Actions and Controls

2.4 Game Rules and Objectives

2.5 Combat and Interaction Systems

2.6 Progression and Reward Systems

2.7 Difficulty and Game Balancing

3. World, Levels and Scene Design

3.1 Game World Structure

3.2 Level Design

3.3 Scene Organization

3.4 Environment Design

3.5 Important Locations

3.6 Objects and Interactive Elements

3.7 Scene Flow and Progression

4. Characters, Assets and Visual Development

4.1 Main Player Character

4.2 Character Design

4.3 Supporting Characters and NPCs

4.4 Enemies and Bosses

4.5 Character Abilities and Attributes

4.6 3D Models and Asset Creation

4.7 Blender Workflow

4.8 Materials and Textures

4.9 Animation and Visual Effects

4.10 Asset Sources and Credits

5. Technical Design and C# Development

5.1 Game Engine and Development Environment

5.2 Project Folder Structure

5.3 Scene and Object Structure

5.4 C# Classes and Objects

5.5 Fields, Properties and Variables

5.6 Methods and Functions

5.7 Conditions and Logical Statements

5.8 Player Input and Behavior

5.9 Object References and Connections

5.10 Gameplay Systems

5.11 Data Management and Game State

6. User Interface and Audio Design

6.1 Main Menu

6.2 Heads-Up Display (HUD)

6.3 Player Information and Status

6.4 Inventory and Equipment

6.5 Settings and Navigation

6.6 Music and Sound Effects

6.7 Environmental Audio

6.8 Accessibility and User Experience

7. Testing, Debugging and Quality Assurance

7.1 Testing Strategy

7.2 Gameplay Testing

7.3 Player Movement and Interaction Tests

7.4 C# Behavior Testing

7.5 Expected Versus Observed Results

7.6 Bug and Error Documentation

7.7 Debugging and Corrections

7.8 Performance Testing

7.9 Final Test Results

8. Documentation and Project Management

8.1 Development Documentation

8.2 Video Learning Records (1–9)

8.3 AI Assistance and Verification

8.4 Team Roles and Responsibilities

8.5 Task Tracking and Ownership

8.6 Discord and Peer Feedback

8.7 Project Demonstration

8.8 Next-Class Development Plan

9. GitHub, Release and Project Handoff

9.1 Project Organization

9.2 Source Code and Asset Management

9.3 Git Version Control

9.4 Commit and Push Records

9.5 Setup and Run Instructions

9.6 Known Issues and Dependencies

9.7 Final Evidence Package

9.8 GitHub Repository Handoff

9.9 Canvas Submission Confirmation

APPENDICES

Appendix A – Whiteboard Planning

Appendix B – Game Flowcharts and Diagrams

Appendix C – Character and Asset References

Appendix D – C# Code References

Appendix E – Screenshots and Development Evidence

Appendix F – Asset Licenses and Source Credits

Appendix G – Glossary of Technical Terms